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Agentic AI in Finance

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Outline

- What is Agentic AI?
 - Applications in Finance
 - Tools used: RAG, LLM

What is Agentic AI?

- Agentic AI refers to artificial intelligence systems that act autonomously, make decisions, and adapt to achieve goals.
- Think of it as a "smart assistant" that:
 - Learns from data
 - Plans and executes tasks
 - Adapts to changing environments



[Nvidia on AgenticAI](#)
[Video link](#)

Benefits of AgenticAI

Benefits of Agentic AI in Finance



Slow reconciliations result in delayed close

Agents reconcile 10× faster with ~99% accuracy → cleaner books



Forecasts go stale quickly

Agents update in real-time → faster treasury action



Finance is buried in manual tasks

Agents handle routine work → teams focus on strategy

- Automated
- Reduce manual/ menial work
- Could be more responsive to environment
- *Less labour...*

Agentic AI vs GenAI

Agentic AI vs Gen AI in Finance

Feature	Generative AI	Agentic AI
Primary Role	✓ Creates content like reports or summaries	✓ Executes finance tasks autonomously
How It Works	✓ Responds to prompts using language models	✓ Uses logic, memory, and reasoning to act on goals
Finance Example	✓ Drafts a budget commentary	✓ Fixes reconciliation issues proactively
Limitations	✓ Needs human input and context	✓ Works independently, with oversight
Best Fit	✓ Report writing, virtual assistants	✓ Cash forecasting, close, anomaly detection

- GenAI is more 'generative' with summaries/ reports (now moving to reasoning).
- AgenticAI is more task-based and can include GenAI as part of its workflow.

What the press says (CNA news)

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Many in Singapore workforce not equipped to handle agentic AI: NTUC Learning

ADVERTIS



AI AT WORK

70 1% of business leaders believe the workforce lacks the skills needed for

0:17 / 3:07

the extent of radioactive contamination at industrial zone South Korea president issues fir

Key Skill for SG Workforce [\(link\)](#)

CISION

Agentic AI Adoption Underway but Singapore Workforce Lacks Familiarity and Skills

PR Newswire Asia

Thu, 2 October 2025 at 2:00 pm SGT • 4 min read



- *Three in four business leaders reveal that their organisations are at various stages of Agentic AI adoption—whether exploratory (23%), testing (34%), or limited deployment within certain departments or functions (18%).*
- *While over three in four business leaders (77%) and nearly three in four employees (74%) agree that adopting and being familiar with Agentic AI is important to remain competitive, almost half of leaders (48%) and more than four in five employees (83%) are unfamiliar with the technology.*
- *Workforce capability is underdeveloped, with 70% of leaders and*

How does the Agentic AI work?

- **Perception:** Agentic AI uses sensors, data inputs, or APIs to gather information from its environment. This could include text, images, numerical data, or real-time feeds from the web or connected systems.
- **Reasoning and Decision-Making:** The AI processes the collected data using advanced algorithms, often powered by large language models (LLMs) or reinforcement learning. It evaluates options, predicts outcomes, and selects actions based on predefined goals or learned objectives.
- **Executes Action:** Once a decision is made, the AI executes actions, which could involve generating responses, manipulating data, controlling devices, or interacting with other systems.
- **Goal-Oriented Behavior:** Unlike traditional AI, which might focus on specific tasks, agentic AI is driven by broader objectives. It can break down complex goals into smaller tasks, prioritize them, and adjust its approach dynamically.

What technologies do the agents use?

- **Machine Learning Models:** LLMs, neural networks, or reinforcement learning frameworks power the AI's reasoning.
- **APIs and Tools:** Agentic AIs often integrate with external systems for data access or action execution (e.g., web search, database queries).
- **Planning Algorithms:** Techniques like hierarchical task networks or chain or feedback loops help the AI break down and pursue goals.
- **Memory Systems:** Short-term and long-term memory (like my ability to recall prior conversations) enable continuity and learning amongst agents. (simpler for LangChain as objects and states live within the key orchestrator)

LLMs in daily use



How do you use LLMs in your daily life?

What if you have your own data set?

- Many LLMs are ‘universal’ – trained on common data
- *What if you have your own dataset?* For e.g. company bespoke information? Or you cannot access the Internet within the organisation?
 - LLMs would not be able to answer specifics as they are not trained on these data.
- Enter:
 - Retrieval-augmented generation (RAG) and
 - Model Context Protocol (MCP)
 - Local server (Ollama)



Customising LLMs.

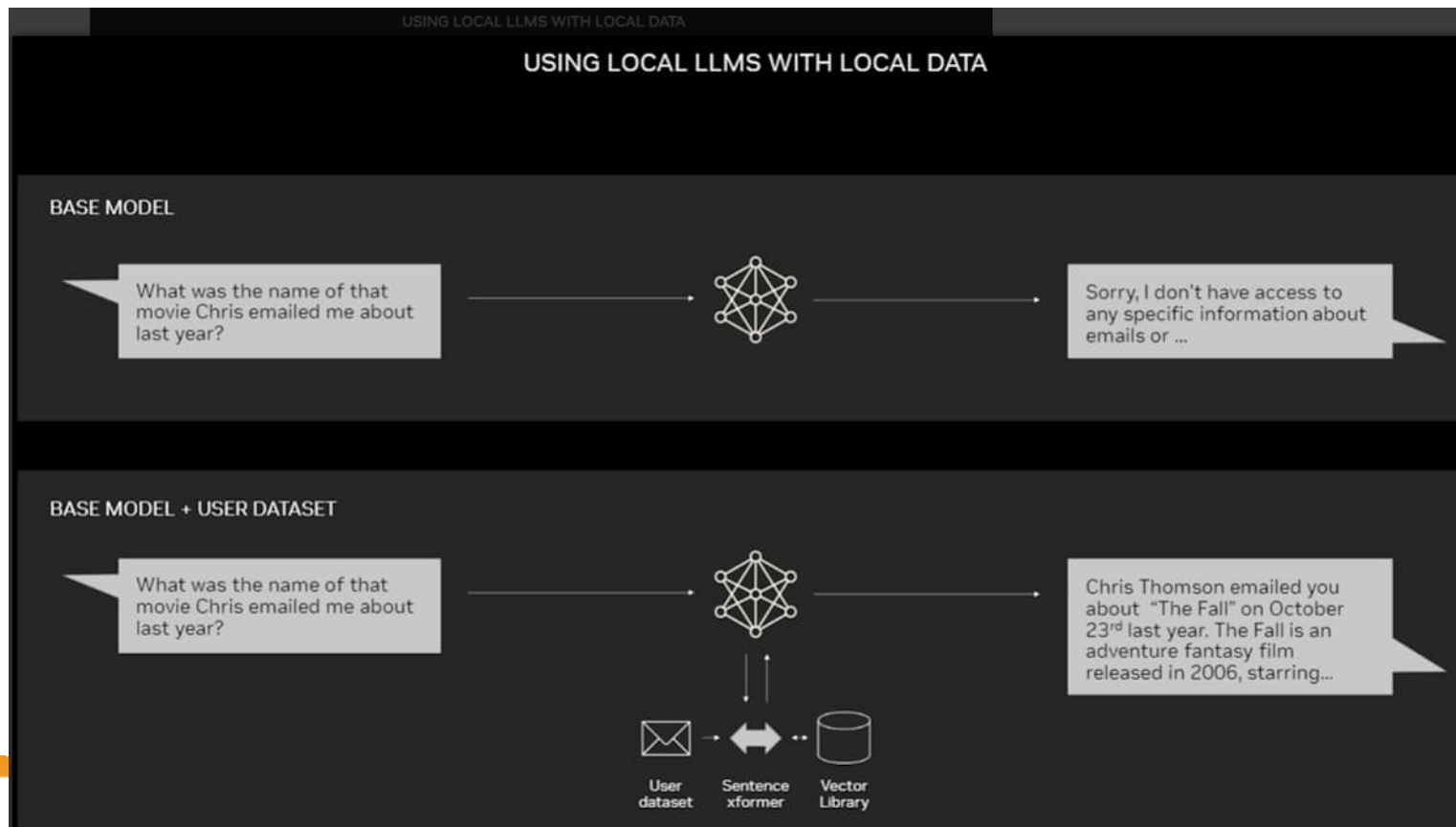
Hosting your own LLMs

- **Ollama** is a **local runtime for large language models (LLMs)** that lets you run modern open-source models **entirely on your own machine** (CPU or GPU), without calling cloud APIs.
- Matters in finance as:

Issue	Ollama's Advantage
Data privacy	Models run locally — no data leaves your machine
Cost control	No per-token API fees
Latency	Millisecond-level local inference
Reproducibility	Fixed model versions
Offline usage	Works without internet
Regulatory comfort	Easier compliance (MAS / GDPR / internal IT)

Retrieval Augmented Generation

- Combines the strengths of large language models (LLMs) with external knowledge bases to ground AI responses to real-world bespoke data and information



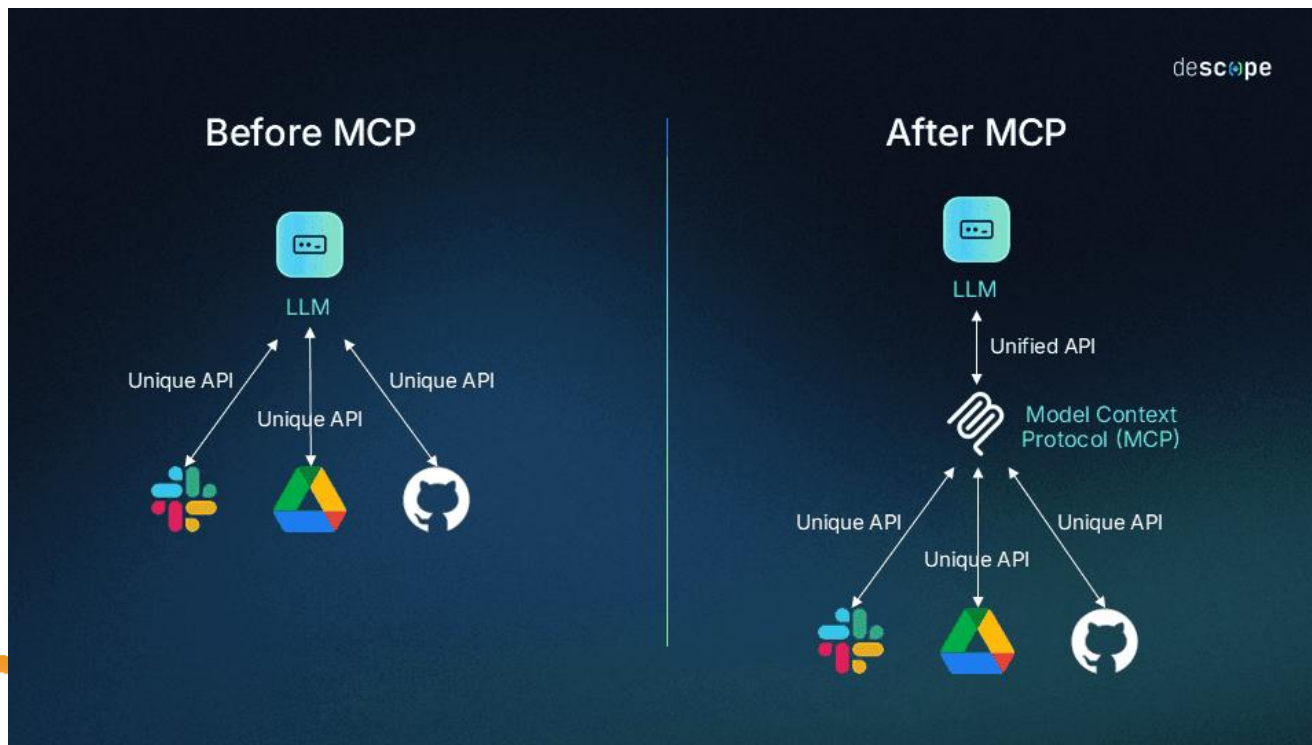
Different RAG Solutions

- Pure document only:
 - Accesses internal documents. These are vectorised into node (bytes) a chromaDB (see e.g.)
 - Useful in finance for eg with Relationship Managers, and research documents.
- Hybrid – document + Web
 - Accesses document (if confidence level or similarity is not high enough) then Internet.
- RAG solutions consist of:
 - Vector database
 - Parsing of documents

Example workbooks provided.

Model Context Protocol (MCP)

- MCP is an open standard that provides a standardized way for large language models (LLMs) to connect to external data, tools, and services. It acts like a universal adapter, similar to a USB-C port for AI, allowing applications to provide LLMs with necessary context to perform tasks, build workflows, and interact with the real world more effectively.



MCP vs RAGs

- RAGs is an architectural pattern while MCP is a standardised protocol.

Aspect	RAG	MCP
Purpose	Ground the model in factual <i>documents</i> or text data	Connect the model to <i>external systems</i> (APIs, databases, analytics tools)
Data Source Examples	Annual reports, PDFs, transcripts, policies, client memos	Market-data feeds, Database feeds
Latency	Slightly slower (retrieval + generation)	Real-time (much faster)
Typical Use Cases	Summarization, Q&A, grounded explanations	Execution, analytics, dynamic querying, system orchestration
Output Nature	Narrative or explanatory text	Data, computations, or structured results
Analogy	“Bring knowledge to the model.”	“Bring the model to the knowledge.”

Exercise: When do you use MCP vs RAGs

Stage

Underlying Method

“What are my largest spending categories this quarter?”

“Explain how my spending pattern compares to last quarter”

“What if I shift \$20,000 from equity to fixed income?”

“Summarize the risk section from my mutual fund’s prospectus”

“Generate a client-ready financial summary PDF”

Enterprise Agentic AI system

There are additional concerns when using agentic AI systems for enterprises.

- Scalability: much larger volume
- Security: protect data sensitivity and compliance with regulations
- Integration: integrate with legacy systems to leverage existing infrastructure
- Robustness: able to operate in variety of environments and conditions
- Customisation: meet specific needs and requirements of different organisation

Enterprise AI Implementation Considerations

Additional considerations with Enterprise Implementation

- RAG with external knowledge base
- Structured output: in JSON for output
- Conversational memory (contextual memory) -> LLMs remember your earlier actions
- AI-powered search engines
- Security and confidentiality of data

Enterprise-ready agents are AI systems specifically designed to meet the needs of businesses and organisations.

Comparison of Agentic AI Frameworks

This course focuses on Ollama, LlamaIndex and LangChain.

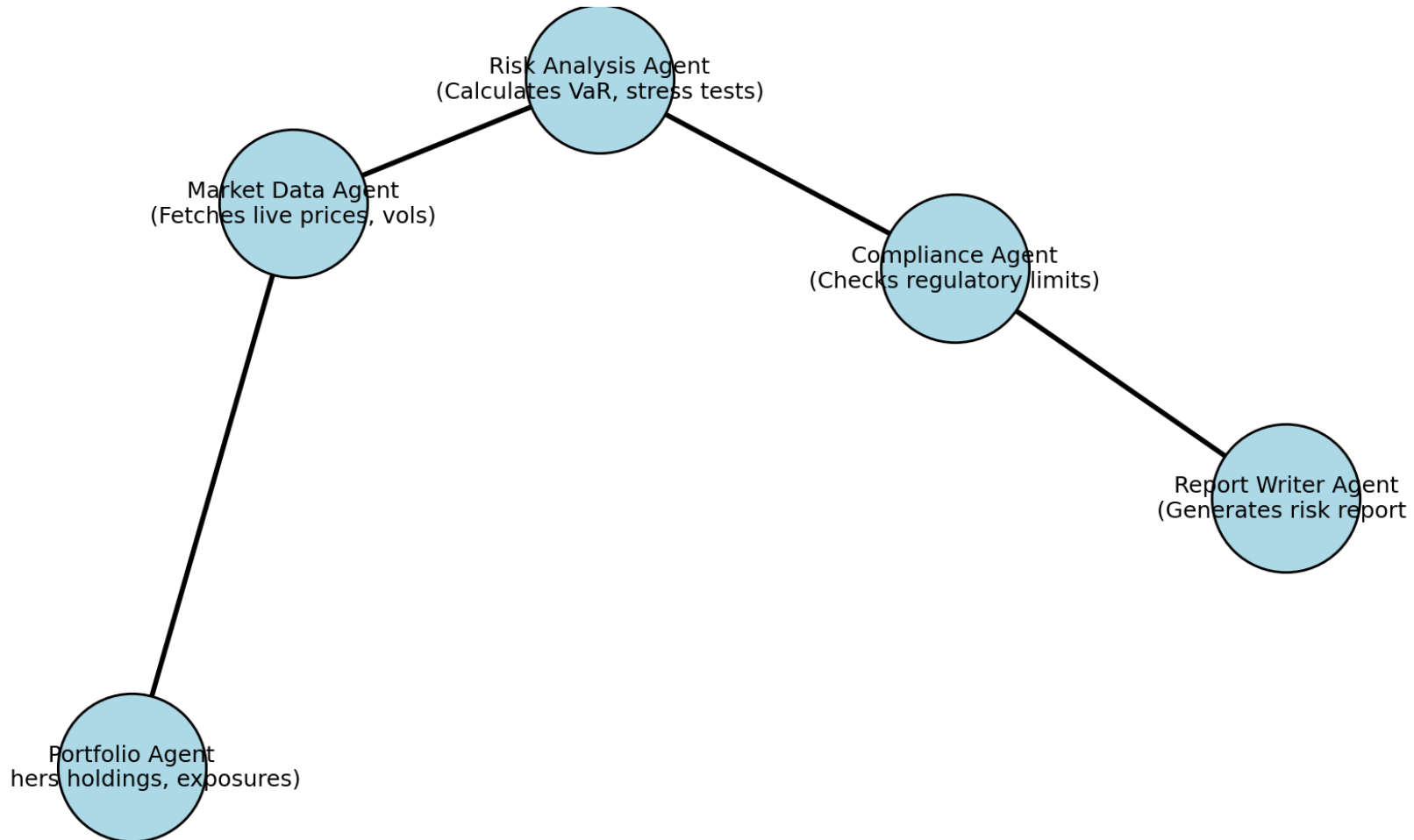
Feature / Purpose	Ollama	LlamaIndex	LangChain	LangGraph
Primary role	LLM runner (local models)	RAG framework	General LLM app framework	Multi-agent workflow orchestrator
Runs models?	✓ Yes (local)	✗ No (connects to APIs/local backends)	✗ No (connects to APIs/local backends)	✗ No (connects to APIs/local backends)
Data connectors / loaders	✗	✓	✓	✗
Vector store / index tools	✗	✓	✓	✗
Agents & tools	✗	✓	✓	✓ (via graph nodes)
Multi-agent orchestration	✗	⚠ Limited	⚠ Limited	✓ Primary strength
Offline capable	✓	✓ (if connected to local model like Ollama)	✓ (if connected to local model like Ollama)	✓ (if connected to local model like Ollama)

Building Blocks of Agentic AI Systems

- **Knowledge Bases**
 - For input + information
 - Model context protocols & RAG
 - For live input and information
- **Large Language Models**
 - For reasoning
- **Actionable APIs**
 - For actioning and execution
- **Orchestration and workflow management**
 - Including supervision and regulation

AgenticAI in Corporate FX Hedging

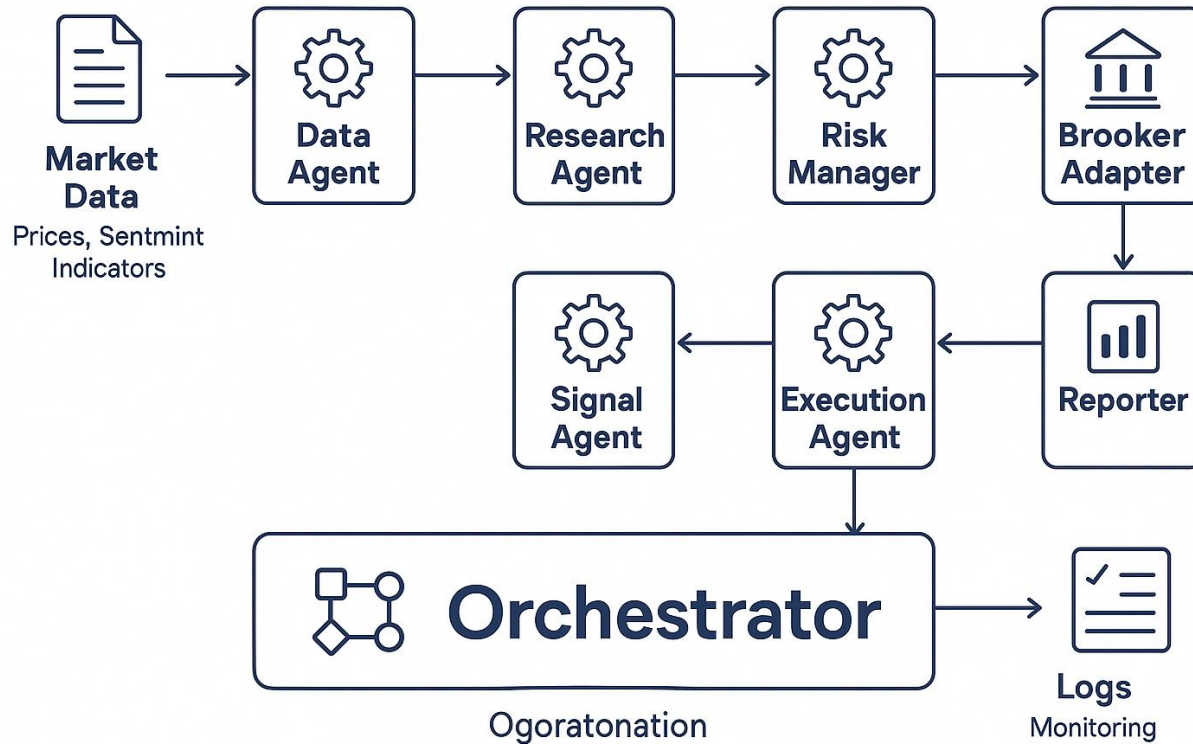
Stateful Multi-Agent Workflow: Corporate Treasury FX Hedging



Corporate FX Hedging Agents

No	Agent	Requires 'Skill'	Function
1	Portfolio agent	Knowledge base	To gather all positions in portfolio
2	Market data	Knowledge base	To keep abreast of market data
3	Risk analysis	Knowledge base, LLM, report writing	To determine risk of portfolio
4	Compliance	Knowledge base, orchestration, actionable APIs	To supervise and limit amount of risks
5	Report writing	LLM	Generate report for management and regulation

AgenticAI in Stock Trading



Stock Trading Agents

No	Agent	Requires 'Skill'	Function
1	Market Data (action)	-	To obtain market data from trusted sources
2	Technical Analysis (action)	Technical indicators etc	Need to make judgement on bullish/bearish etc.
3	Risk management (perception)	Risk of existing portfolio, exposure limits etc.	Make judgement on key market risks in trading
4	Execution (supervision)	Final decision point/judgement	Executes trade to broker

Agents do either action, perception or supervision.

Case Study: Design Workflow in Agentic AI

- **Define Objectives:** Clearly state what the agentic workflow needs to accomplish.
 - **Map Current Process:** Analyze the existing process to identify tasks and potential areas for automation.
 - What patterns are appropriate?
- Determine the **types of processes** encompass: perception, action, supervision in each agent?
- What **memory** is needed? To pass between agents?
- What is the **knowledge base**? Data input.. Either dynamic (by MCP) or static?
- **What LLM** would you use? Is it for coding? Or writing? Etc.
- What **type of actions** are needed? Send emails/ reports?
- What type of supervision? Human in the loop?

Case Study: Work Design for Agentic AI

- Discuss in your job scope the possibilities to use AgenticAI.
- Frame the job into various steps in a work design stage.
- Use as guidelines the design workflows and also the describe the use/ type of agents you would use at each stage.
- **Human judgement, adaptability and ethics are KEY.**

The findings highlight that there is still a fundamental lack of understanding of AI and Agentic AI across workplaces. While adoption has started taking place, many employees and leaders themselves admit they are unfamiliar with the technology, and unsure of its implications. However, the success in this new era will take more than just technical know-how. Technical skills are essential, but human judgement, adaptability and ethics remain indispensable. Redesigning current jobs and developing the human capabilities enable workers to integrate Agentic AI thoughtfully and successfully in their day-to-day roles. By building this dual

Conclusion

Concept & Evolution:

Agentic AI refers to autonomous, goal-directed systems that perceive, reason, and act using data, APIs, and planning algorithms that moves from rule-based automation to adaptive, intelligent decision-making.

Enterprise & Technical Foundations:

Enterprise-ready agentic systems combine LLMs, machine-learning models, APIs, memory, and orchestration tools while ensuring scalability, integration, and security within corporate environments.

Workflow & Design Patterns:

Common patterns include delegation/hand-off, tool-use and function-calling, sequential chains, judge-and-critic feedback loops, parallelisation for speed, and guardrails for safety, governance, and privacy.

Finance Applications & Case Workflows:

Demonstrated through corporate FX hedging and stock-trading agents, the slides emphasise designing workflows that define objectives, map processes, choose agent types, set memory and knowledge bases, specify actions and determine appropriate human supervision.